

Signal Detection in High-Noise Time Series Data Using R

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Introduction

- There are many challenges in detecting signals within noisy data.
- The main goal is to enhance the detection of stationary sinusoidal signals within noisy time series data using R software.
- This methodology can be applied in telecommunications, seismology, economic forecasting.

Spectral Leakage

- Spectral leakage occurs when the signal is not perfectly periodic within the observed time window, leading to a dispersion of energy across adjacent frequencies.
- Spectral leakage yields to pseudo-spectral interpretation by masking the true frequency components and significant features of the signal.

Experimental design

Two Paths:

- Enhanced spectral analysis methodology vs.
- Standard spectral analysis packages.

Criteria:

- Frequency detection within a 20% tolerance.

Comparison for different packages in R

- Purpose:

Assess performance in frequency detection and noise.

- Packages Used:

TSA, astda, spectral, signal, seewave, psd, multitaper, kza.

Methodology

- Objective:

Enhance signal detection in noisy environments using R.

Techniques:

1. zero padding
2. dataset repetition
3. Kolmogorov-Zurbenko filter
4. Spectral differentiation

Kolmogorov - Zurbenko filter (KZ)

$$Y_t = \frac{1}{m} \sum_{j=-k}^k (X_{t+j})$$

- KZ-Filter equation utilized for the moving average adapted from Zurbenko (1986); package kza (Close et al. 2020).

Function:

- Reduces short-term fluctuations and outliers.
- Enhances pattern and frequency identification.

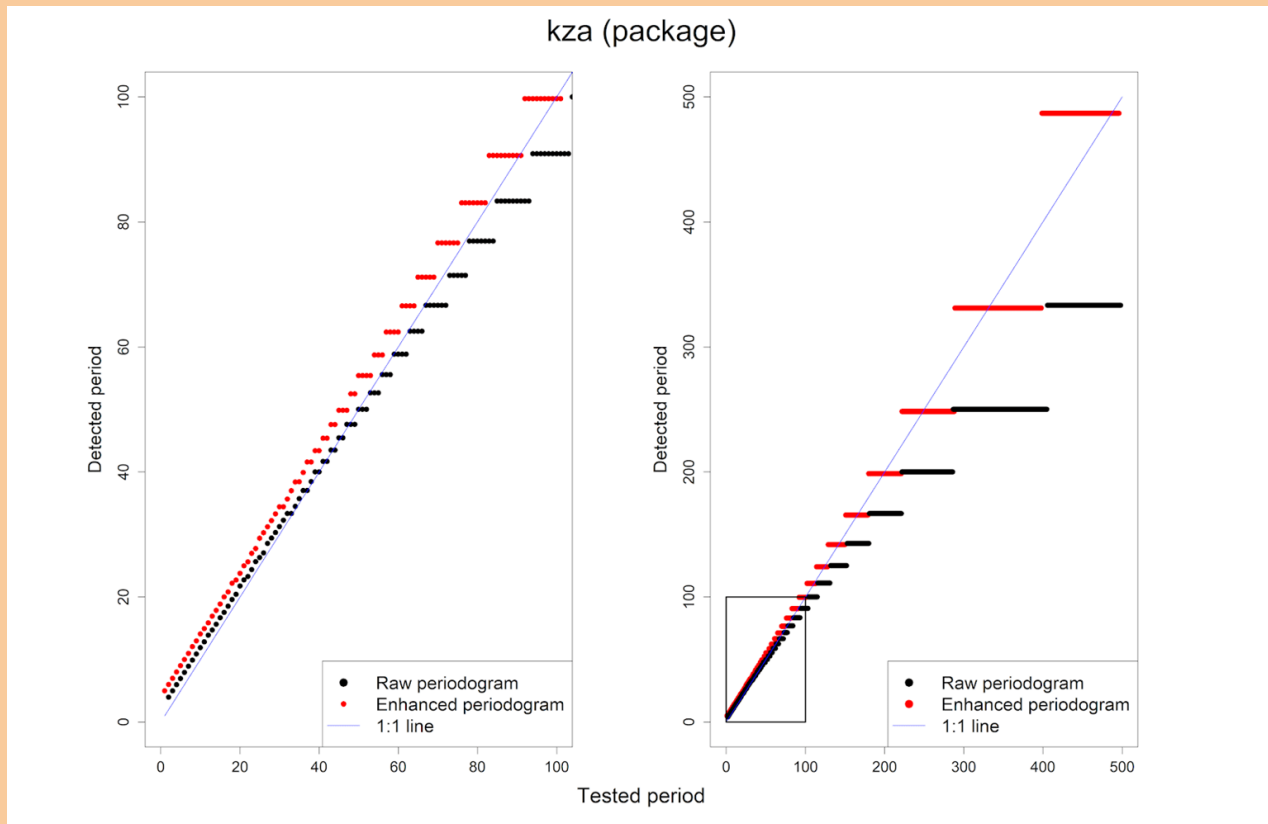
Synthetic data

- Sinusoidal wave within 1000-point sample space.
- Superimposed with random noise varying from 1 to 20 times the signal magnitude.

Results: signal detection comparison

Comparative Analysis:
Enhanced methodology
vs. standard packages
across noise levels.

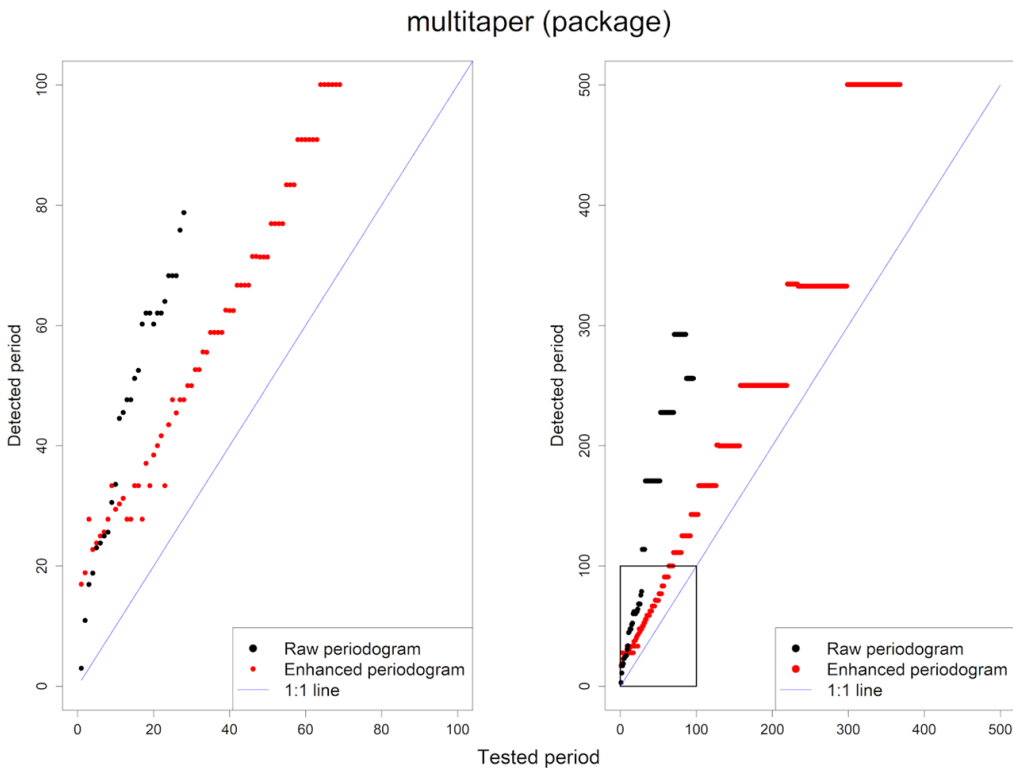
Performance:
Significant
improvement in
frequency detection
with the enhanced
methodology.



Results: signal detection comparison

Performance:

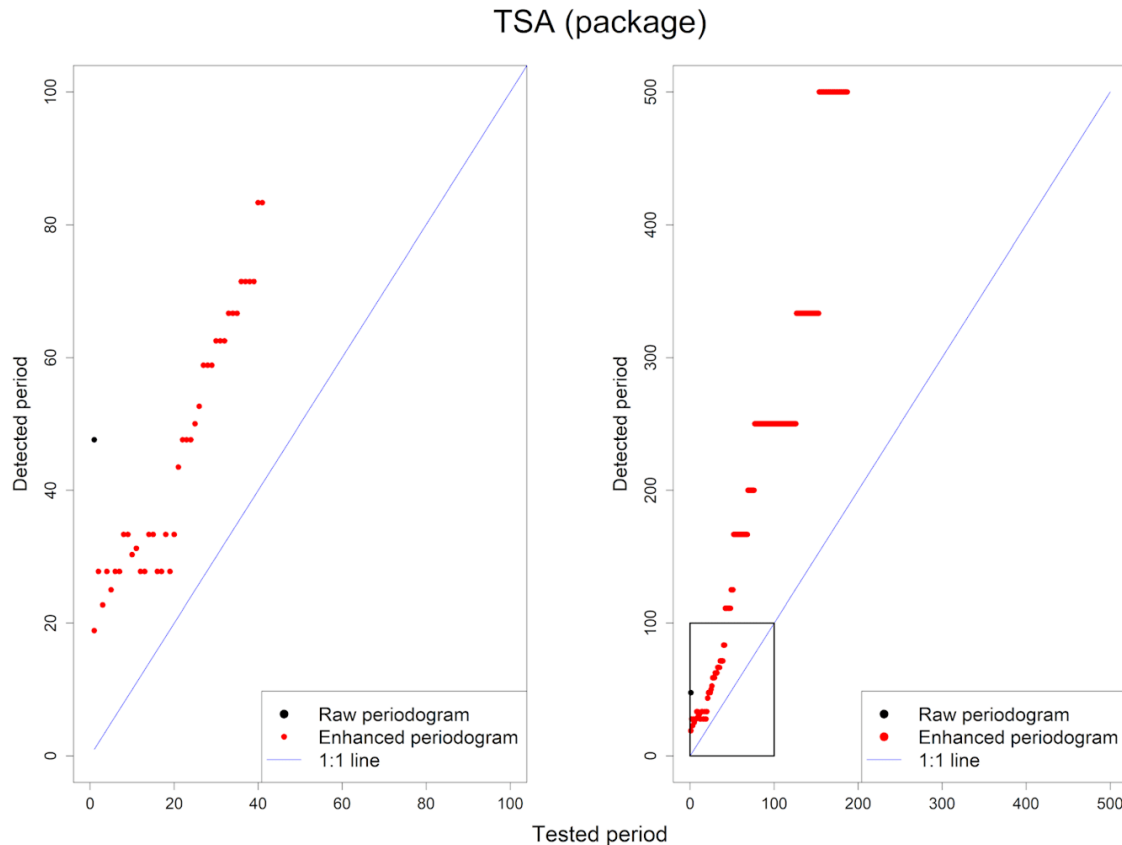
Red dots (enhanced method) align closer to true frequencies than black dots (standard method).



Results: signal detection comparison

Performance:

Enhanced method
outperforms
standard method,
especially under
high noise
conditions.



Conclusions

- Enhanced methodology significantly reduces spectral leakage.
- Improved signal detection even under challenging noise conditions.
- Enhanced methodology outperforms standard packages in noisy environments.
- Potential for broader applications in various fields requiring precise signal analysis.

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Spectral differentiation and zero padding

- **Spectral differentiation**

Technique:

- Elevates spectral density values to the fifth power, subtracts the fourth power.

Result:

- Enhances dominant peaks, improves signal-to-noise ratio.

- **Zero padding**

Technique:

- Extends signal length without altering content.

Result:

- Improves frequency resolution in FFT-based spectral analysis.